

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	1 June 2026
Team ID	
Project Name	ShopEZ: E-Commerce Application
Maximum Marks	4 Marks

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through form using name, email, and password
FR-1	User Registration	Validate required fields during registration
FR-1	User Registration	Store registered user details in database
FR-2	User Login	Login using email and password
FR-2	User Login	Validate user credentials
FR-2	User Login	Display error message for invalid login
FR-3	User Authentication	Maintain secure user session after login
FR-3	User Authentication	Allow logout from the application
FR-3	User Authentication	Restrict protected pages from unauthorized users
FR-4	Product Browsing	View list of available products
FR-4	Product Browsing	View product details such as name, image, price, and description
FR-4	Product Browsing	Search or filter products

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-5	Store Management	View available stores
FR-5	Store Management	Add new store details
FR-5	Store Management	Update existing store details
FR-5	Store Management	Delete store details
FR-6	Product Management	Add new product
FR-6	Product Management	Update product details
FR-6	Product Management	Delete product
FR-7	Cart Management	Add product to cart
FR-7	Cart Management	View cart items
FR-7	Cart Management	Update product quantity in cart
FR-7	Cart Management	Remove product from cart
FR-7	Cart Management	Calculate total cart amount
FR-8	Order Management	Place order from cart
FR-8	Order Management	Store order details in database
FR-8	Order Management	View order confirmation
FR-8	Order Management	View order history

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-9	Admin/Seller Management	Admin/seller can manage products, stores, and orders
FR-9	Admin/Seller Management	Admin can view registered users
FR-9	Admin/Seller Management	Admin can update order status
FR-10	Database Management	Save user, product, store, cart, and order data
FR-10	Database Management	Retrieve, update, and delete application data

Non-functional Requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should have a simple and user-friendly interface so customers can browse products, manage carts, and place orders easily.
NFR-2	Security	The application should protect user data using authentication, password hashing, protected routes, input validation, and secure database connection.
NFR-3	Reliability	The system should work consistently without data loss and should correctly store user, product, cart, and order information.
NFR-4	Performance	The application should load pages and API responses quickly. Product lists, cart updates, and order processing should respond within acceptable time.

FR No.	Non-Functional Requirement	Description
NFR-5	Availability	The application should be accessible whenever users need it. If deployed online, frontend, backend, and database services should remain available.
NFR-6	Scalability	The system should support future growth, including more users, products, stores, and orders by scaling frontend, backend, and database layers.